



Create Django Project & Run Server

Lecture #4: django-admin startproject • manage.py • Architecture Tree • Port 8000

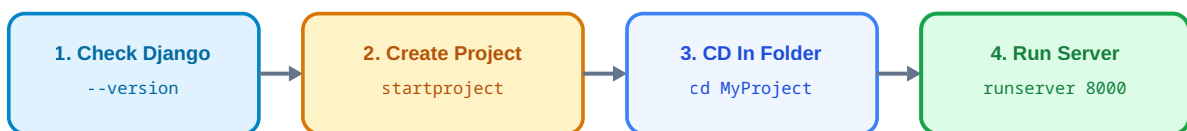
Colorful Handbook • Mohit Decodes Hindi Series #4 • Easy Hinglish

🎯 Quick Connection (Python & Database Synergy):

Pichle lectures me humne theory, MVT architecture aur Virtual Environment dekha tha. Aaj hum practically pehla live local server start kar rahe hain! Aapka Python CLI command flow aur aane wale database configurations ka pura base yahi se create hota hai.

🌍 1. Project Creation & Run Server Pipeline

Naya project banane aur local server live karne ke 4 aasan steps:



2. Step-by-Step Terminal Commands (Code Snippets)

Step 1: Django Check Karein

```
django-admin --version  
# Output e.g.: 5.2.4 (Agar version dikhe toh aap project banane ke liye ready hain!)
```

Step 2: Naya Project Banayein

```
django-admin startproject MyProject
```

Ye command run hote hi aapke folder ke andar `MyProject` naam ka folder aur Django ka poora scaffolding code generate ho jata hai.

Step 3: Project Folder Ke Andar Jaayein

```
cd MyProject
```

Sabse Common Beginner Galti:

`manage.py` file hamesha project ke root folder ke andar hoti hai. Agar aap `cd MyProject` kiye bina bahar se hi `runserver` chalayenge toh `python: can't open file 'manage.py': [Errno 2] No such file or directory` ka error aayega!

Step 4: Development Server Run Karein

```
# Windows (CMD / PowerShell):  
python manage.py runserver  
  
# macOS / Linux:  
python3 manage.py runserver
```

Server Output & Success Verification:

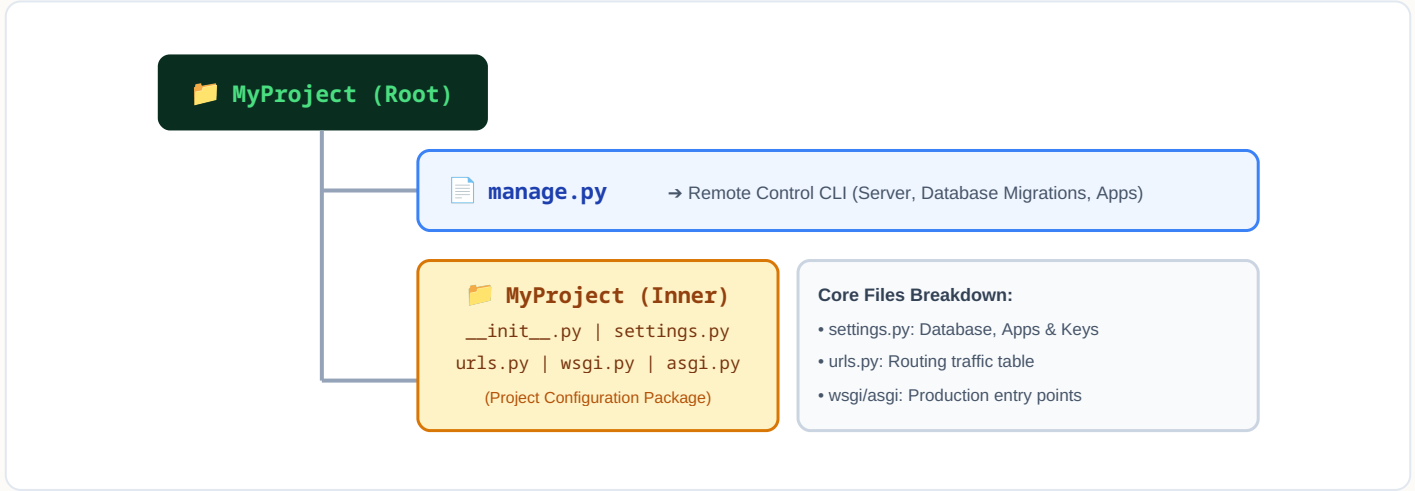
Terminal me message aayega:

```
Starting development server at http://127.0.0.1:8000/
```

Browser me `http://127.0.0.1:8000` ya `localhost:8000` open karte hi Django ka official animated rocket aur *"The install worked successfully! Congratulations!"* ka welcome page dikhega!

3. Django Project File Structure (Naksha & Roles)

Jab aap `startproject` run karte hain, toh Django niche diya gaya structure banata hai:



File / Directory	Kaam aur Asli Role (Simple Hinglish)
<code>manage.py</code>	Django Ka Remote Control! Saari commands (server chalana, migration karna, nayi app banana) isi script se run hoti hain. Isko manually kabhi edit nahi kiya jata.
<code>__init__.py</code>	Ye empty file hoti hai jo Python ko signal deti hai ki ye directory ek <i>Python Package</i> hai jiske andar ke files/modules ko import kiya ja sake.
<code>settings.py</code>	Project Ka Control Room / Dimaag! Database configurations (SQLite, PostgreSQL, MySQL), installed apps, security keys, static files sab yahi configure hoti hain.
<code>urls.py</code>	Address Book / Traffic Police! User jo bhi URL request karega, ye file use uske matching View function tak route karti hai.
<code>wsgi.py</code> / <code>asgi.py</code>	Production Deployment Gateways! Live production servers (Nginx, Gunicorn) ko Django web app se connect karne ke liye use hoti hain.

-  **Corporate Office Analogy (Yaad Rakhna Super Easy):**
- `manage.py` = **Office Manager:** Jo subah office ka shutter kholta hai (Server on) aur saare staff ke tasks coordinate karta hai.
 - `settings.py` = **Rulebook & Accounts:** Office ke saare niyam, passwords aur database credentials store karta hai.
 - `urls.py` = **Receptionist:** Visitor ke aate hi usko sahi cabin ya meeting room (View) bhejti hai.
 - `wsgi.py` = **Main Entrance Gate:** Bahar se aane wale Internet traffic ko office ke andar laata hai.

4. Server Controls & Important Commands

🛑 Server Band / Stop Karna

Jab server chal raha ho aur aapko terminal me naya command likhna ho ya server quit karna ho:

Press: `Ctrl + C`

Isse development server safely stop ho jata hai.

🔧 Custom Port Par Chalana

By default Django **Port 8000** par chalta hai. Agar port busy ho ya dusre port par chalana ho:

```
python manage.py runserver 8080
```

Ab server `http://127.0.0.1:8080/` par chalega.

💡 5. Top 5 Interview / Exam Quick Revision Q&A

1. Q: Naya Django project create karne ka command kya hai?

Ans: `django-admin startproject <project_name>`

2. Q: Development server start karne ka command kya hota hai?

Ans: `python manage.py runserver` (macOS: `python3 manage.py runserver`)

3. Q: Django ka default development port aur IP kya hota hai?

Ans: IP: `127.0.0.1` (Localhost) aur Port: `8000`.

4. Q: `manage.py` aur `settings.py` me kya antar hai?

Ans: `manage.py` CLI utility hai commands execute karne ke liye, jabki `settings.py` project configurations aur database credentials store karti hai.

5. Q: `__init__.py` file empty hone ke bawajood directory me kyun hoti hai?

Ans: Python interpreter ko ye inform karne ke liye ki ye directory ek importable *Python Package* hai.